Portfolio Risk

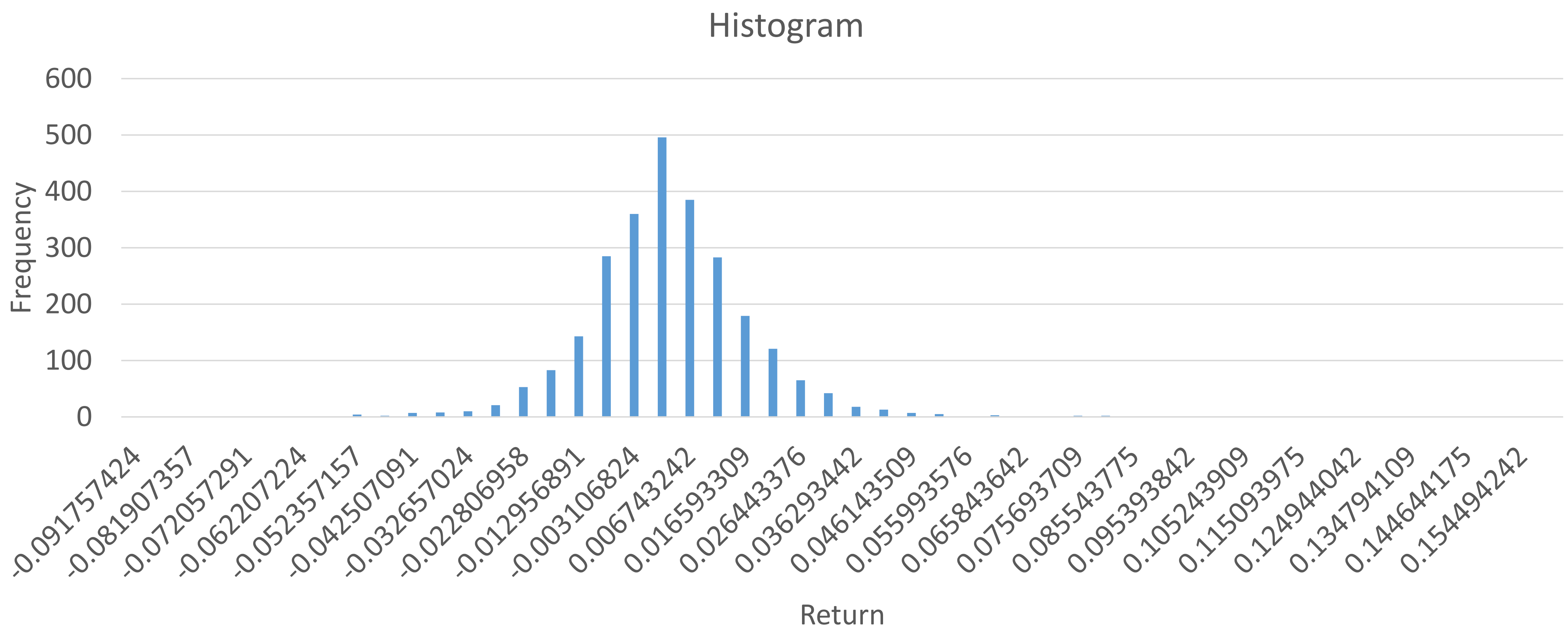
Portfolio Risk Exposures

- Sensitivity measures
  - Deviation of market prices due to a unit change of a specific market parameter
- Volatility measures
  - Variations around the average market price
- Downside measures of portfolio risk
  - Focuses only on the losses incurred by portfolios

Portfolio VaR Measurement

- Value at Risk (VaR) was developed in 1993.
- Purpose of VaR is to quantify financial risks using standard statistical techniques.
- VaR measures the worst expected loss over a given horizon under normal market conditions at a given confidence level.
- Estimated level of loss on an asset or portfolio for a specified probability (confidence level) and time horizon
- VaRα(X) is a lower α-percentile of the random variable X

Return Distribution



Requirement for VaR Measurement

- Amount of exposure
  - mark-to-market value of the asset or portfolio
- Risk factor or factors
  - source of variability of the market value of the asset or portfolio (price, reference rate, index)
- Risk horizon
  - length of the risk period has to exceed the liquidation time of the asset or portfolio
- Data series of the risk factors
  - frequency of the data series equals the risk horizon for a long period
- Level of confidence
  - confidence levels from 90 - 99.9 percent

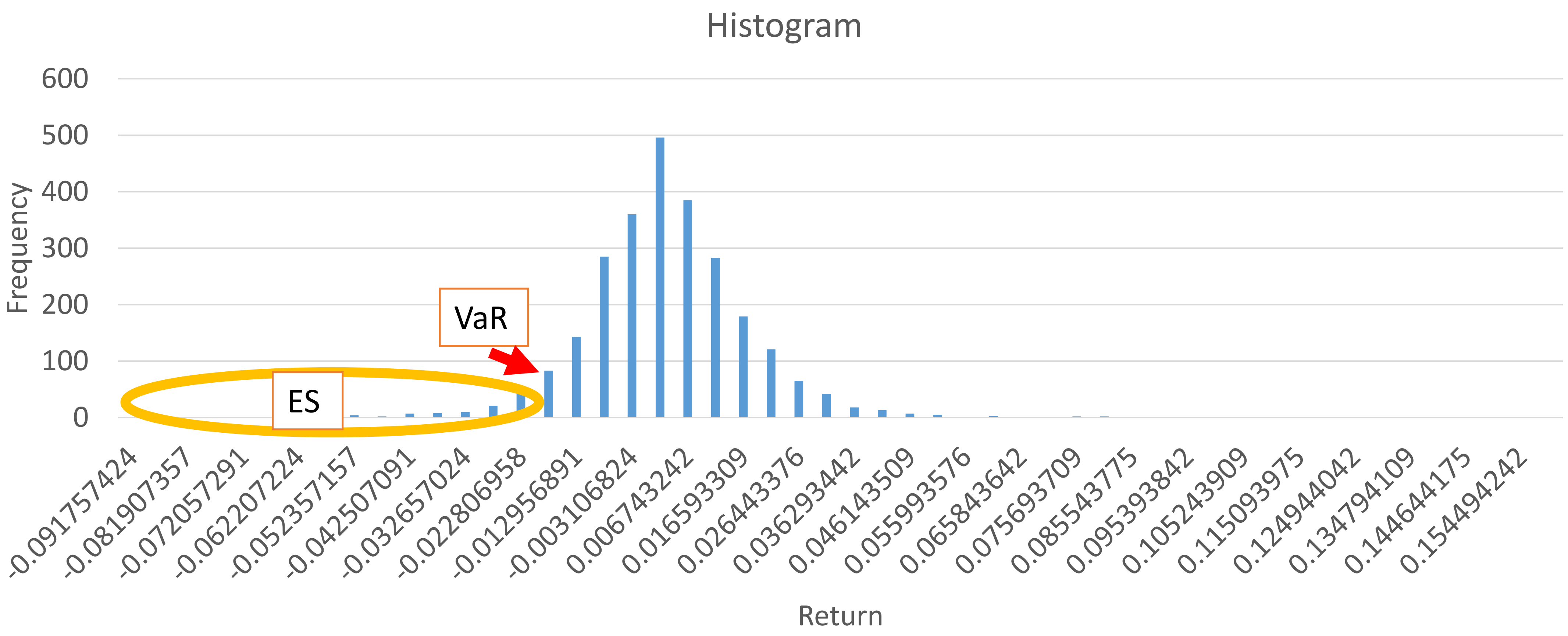
Measure of VaR

- Parametric
- Parametric methods assume a normal distribution
- Nonparametric (historical simulation)
- Nonparametric methods make no assumption regarding the distribution.
- Monte Carlo simulation
- Monte Carlo method simulates multiple random scenarios

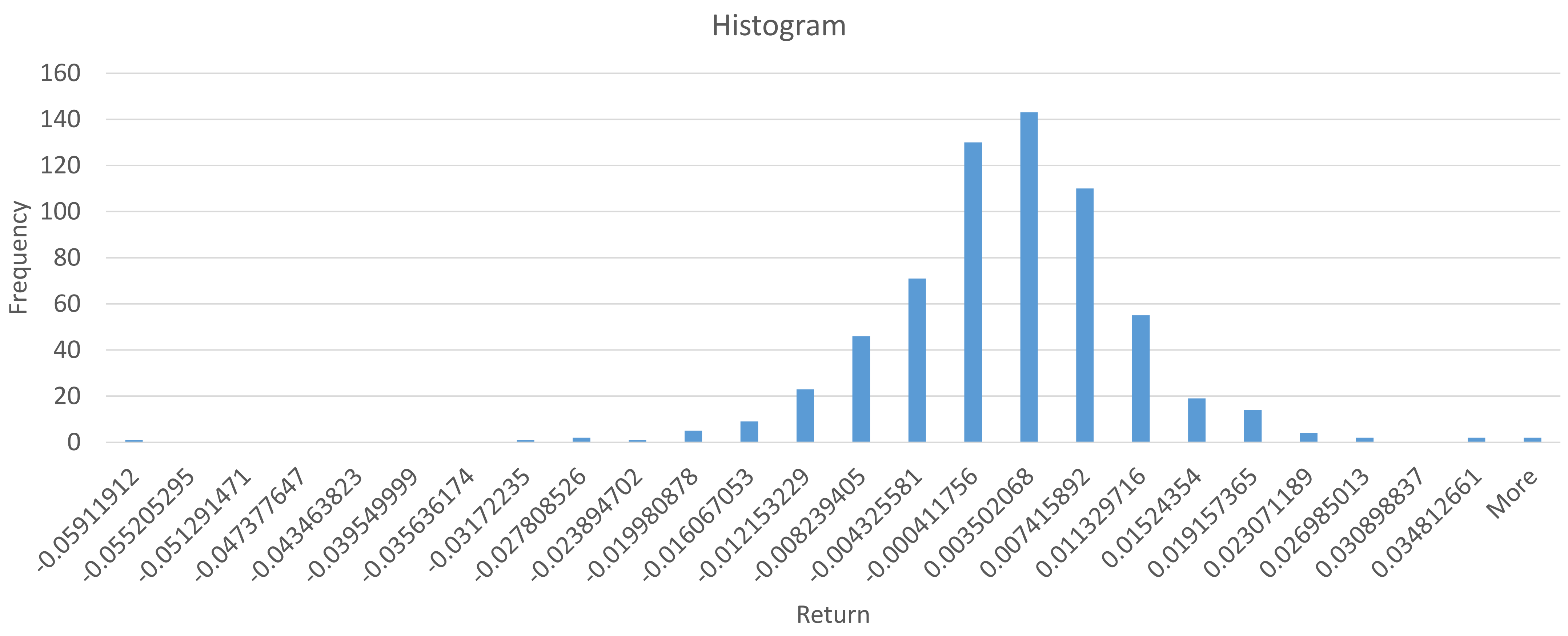
Expected Shortfall

- VaR has tail risk when VaR fails to summarize the relative choice between portfolios as a result of its underestimation of the risk of portfolios with fat-tailed properties and a high potential for large losses.
- Tail risk of VaR emerges since it measures only a single quantile of the profit/loss distributions and disregards any loss beyond the VaR level.
- VaR at the 95% confidence level of portfolio Sensex is 14,000 and that of portfolio Nifty is 15,000. Given these numbers, one may conclude that portfolio Nifty is more risky than portfolio Sensex. However, the investor does not know how much may be lost outside of the confidence interval when the maximum loss of each portfolio is considered.
- Expected shortfall has tail risk when expected shortfall fails to summarize the relative choice between portfolios as a result of its underestimation of the risk of portfolios with fat-tailed properties and a high potential for large losses.

Return Distribution



Historical Returns Sensex



Parametric Method

- Significance level α : portfolio σ

$$VaR_{portfolio} = \alpha \times \sigma_{portfolio}$$

	Value
Initial Portfolio value	1000000
Number of observations: Sensex	640
Parameter: Standard deviation	0.008679
Parameter: Average	0.000121
Confidence Level:	95%
VaR	VaR_95 = norm.ppf(0.05, mean, std_dev)
VaR	**-14154.9**

Historical Method

- Sort the portfolio returns in ascending order to arrive at an observation distribution of changes in portfolio value.
- VaR number will be equal to that percentile associated with the specified level of confidence.
- k^{th} percentile means that the lowest k per cent of the sample of changes in portfolio value will exceed the VaR measure. For a 95 per cent level of confidence, the VaR number equals the 5^{th} percentile of the distribution of changes in portfolio value.
- When there are 650 observations, this implies 32.5 losses (5 per cent of the sample) will be larger than the VaR measure (VaR measure is essentially the 32.5 lowest observation).

Historical Simulation

- Historical simulation is achieved by selecting the data set and drawing sub samples from the data set.
- Computation of VaR by choosing the 95th percentile of the sub sample when the data is arranged in a rank order, a probability distribution of VaR is created.
- Next step is to create random numbers using the rand function in a spreadsheet application.
- Results in simulated random numbers.
- Historical VaR and assigned probability is then used to establish VaR from the simulated sample.

Example

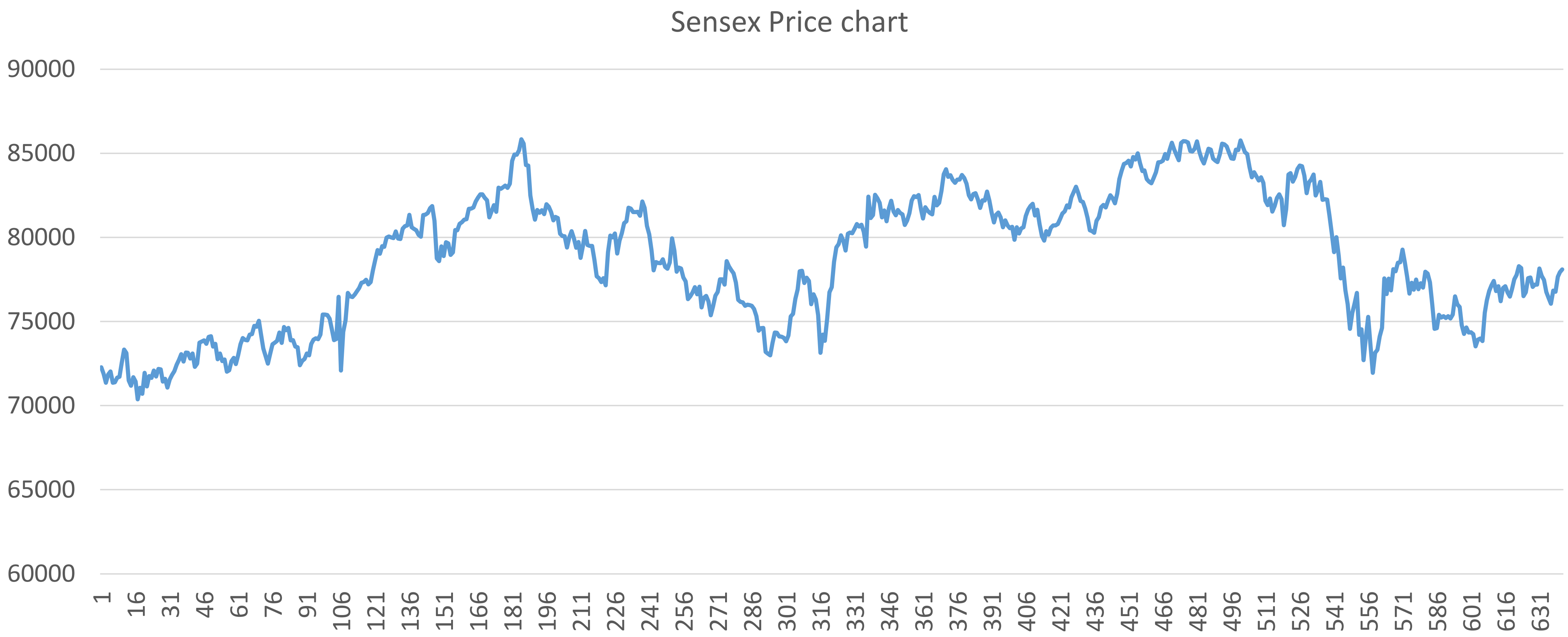
Historical Return %	Historical VaR	Historical Probability	Historical Cumulative Probability
5	-2.00	0.05	0.05
15	-1.80	0.15	0.2
20	-1.60	0.2	0.4
2	-1.40	0.02	0.42
25	-1.20	0.25	0.67
15	-1.00	0.15	0.82
3	-0.80	0.03	0.85
10	-0.60	0.1	0.95
5	-0.40	0.05	1
100			

- Cumulative probability measures are used to convert the simulated random returns to VaR measures.
- Cumulative probability of -2% losses is 0.05.
- When the random number generated by the simulation function is within the 0.05 range then the VaR of -2% is assigned.
- If the generated random number is greater than 0.05 but within 0.15, then VaR of -1.8% is assigned.
- Similarly the entire simulated series is assigned a random VaR based on the historical probability values.
- VaR at 95th percentile is then chosen as portfolio VaR through the historical simulation after the ranking of random VaRs.
- VaR at 95% percentile is estimated as -1.8%.

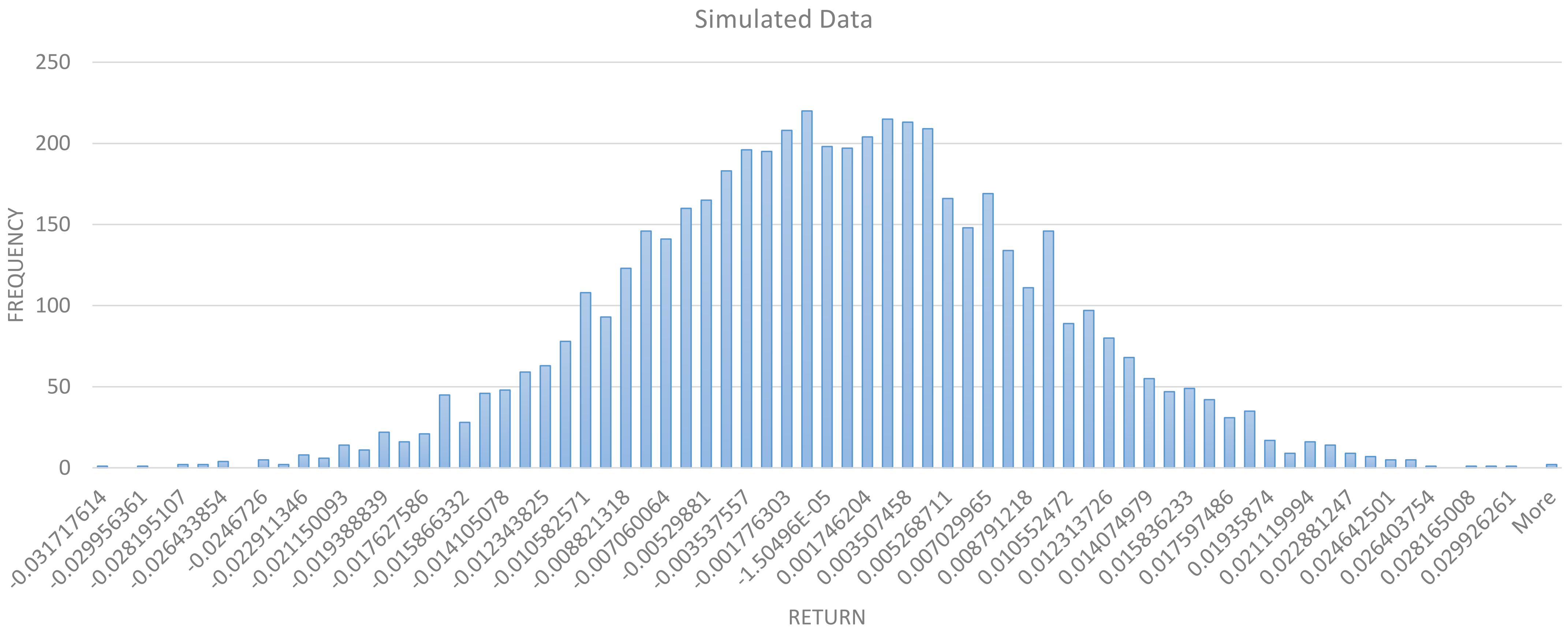
Monte Carlo Simulation

- First step in Monte Carlo simulation is to identify the market risks that are likely to change the portfolio value either on an overall basis or in terms of individual security components.
- Factors leading to a change in portfolio value are then simulated wherein the probability distribution representing each market risk factor has to be identified.
- Scenarios thus depict the portfolio change due to each market risk factor with an assumed probability distribution.

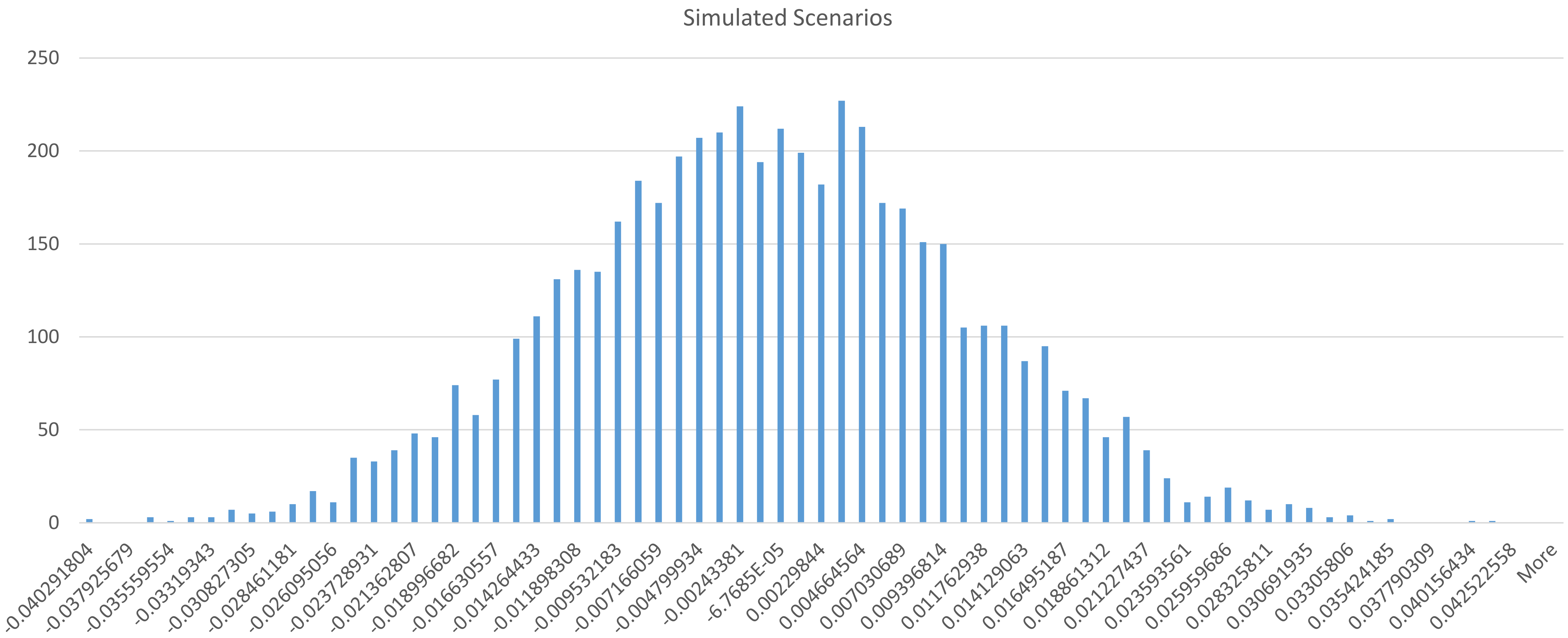
Price Trend Scenarios



Simulation Method (Sensex)



Simulation Method



Historical VaR & Monte Carlo VaR

	Value
Initial portfolio value	1000000
Number of observations	640
Number of simuations	5211
Lower 5% of observations	260
Average return	-4.17E-05
Standard deviation	0.008763
5% daily VaR return	-0.0072371
5% Var Value (Monte Carlo)	**-7237.1193**
5% daily VaR return	-0.0072444
5% Var Value (Historical)	**-7244.4426**
Expected Shortfall	0.0121771

Evaluation of VaR

- Advantage of using Monte Carlo simulation to compute value at risk are many when compared with that of variance co-variance and historical simulation measures of VaR computation.
- Variance co-variance approach assumes that the portfolio returns are normally distributed which is very difficult to satisfy in real market conditions.
- Since VaR is a measure of downside risk, most often it is the left side of the distribution which is to be considered for determination of portfolio risk.
- Computation of VaR hence through Monte Carlo VaR is better since the exact distribution that is fit on factors behind portfolio return variations are mapped individually and not as a overall probability distribution.

Marginal VaR

- Marginal VaR is defined as the partial derivative with respect to the component weight.
- It measures the change in portfolio VaR resulting from adding additional value (in one unit) to a component.

$$\Delta VaR_i = \frac{\alpha \sigma_{ip}}{\sigma_p} \qquad \Delta VaR = \alpha \beta \sigma_p$$

Marginal VaR

- Marginal VaR = dollar Covariance / Portfolio Volatility * deviate
- Marginal VaR = Total portfolio VaR / Weight * beta(i,P) (Deviate * dollar portfolio volatility * beta(I,P))
- Beta = Covariance (i,P) / Variance(P)
- Beta = W * Dollar Covariance (i,P) / Dollar Variance(P)
- Dollar Covariance = sigma * x = Covariance Matrix * Position Vector

Component VaR

- Component VaR is a partition of the portfolio VaR that indicates the change of VaR if a given component was deleted.
- Component VaR is used to have a risk decomposition of the current portfolio.

$$\text{Component } VaR = (\Delta VaR_i) \times w_i P = VaR \beta_i w_i \,.$$

Incremental VaR

- Incremental VaR measures the change in VaR due to a new position on the portfolio.

$$\text{Incremental } VaR = VaR_{p+a} - VaR_p \, .$$